

Task 5 — Exploratory Data Analysis (EDA) on Titanic Dataset

Data Analyst Internship — Elevate Labs

Objective

Extract meaningful insights from the Titanic dataset using visual and statistical exploration techniques — covering univariate, bivariate, and multivariate analysis.

Tools & Libraries

Tool	Purpose
Python 3	Core language
Pandas	Data loading, cleaning, aggregation
NumPy	Numerical operations & log transforms
Matplotlib	Base plotting
Seaborn	Statistical visualizations (heatmap, pairplot, boxplot)
Jupyter Notebook	Interactive analysis environment

Repository Structure

```
task-5-titanic-eda/
|
├── Titanic_EDA.ipynb           # Main Jupyter Notebook with all code +
observations
├── Titanic_EDA_Report.docx     # PDF-ready findings report
|
├── train.csv                  # Training dataset (891 rows × 12 columns)
├── test.csv                   # Test dataset
├── gender_submission.csv      # Sample submission file
|
└── README.md                  # This file
```

What Was Done

1. Dataset Overview

- Loaded `train.csv` (891 rows, 12 columns) using `pd.read_csv()`
- Explored column types and descriptions using `.info()` and `.describe()`

2. Missing Value Analysis

- Identified 3 columns with missing data using `.isnull().sum()`
- Cabin:** 77.1% missing → too sparse to use directly
- Age:** 19.87% missing → recommended median/group imputation
- Embarked:** 0.22% missing → filled with mode ('S')

3. Univariate Analysis

- Plotted survival distribution (count + pie chart) — 38.4% survived
- `.value_counts()` on categorical features: Pclass, Sex, Embarked
- Histograms for numeric features: Age, Fare, SibSp, Parch

4. Bivariate Analysis

- Survival rate grouped by Pclass, Sex, Embarked (bar charts)
- Boxplots of Age & Fare by Survival status
- Scatterplot of Age vs Fare coloured by Survival

5. Multivariate Analysis

- Correlation heatmap (lower-triangle, `sns.heatmap()`)
- Pairplot with KDE diagonals, hued by Survived (`sns.pairplot()`)
- Pivot heatmap: Survival Rate by Pclass × Sex

6. Skewness & Outlier Detection

- Computed skewness for all numeric features
- Demonstrated `log1p` transform on Fare (skew: 4.79 → near-normal)
- Boxplots for outlier detection on Age, Fare, SibSp, Parch

7. Feature Engineering

- Created `FamilySize = SibSp + Parch + 1`
 - Created `IsAlone = (FamilySize == 1)`
 - Plotted survival rates by FamilySize and IsAlone
-

🔑 Key Findings

#	Finding	Metric
1	Overall survival rate is low	38.4% survived
2	Gender is the strongest predictor	Female: 74% vs Male: 19%
3	Passenger class is critical	1st: 63% vs 3rd: 24%
4	Fare positively correlates with survival	Correlation = +0.26
5	Children were prioritised	Ages 0-12: 59.4% survival
6	Travelling with family improves survival	Solo: 30% vs Family: 51%
7	Cabin data is too sparse to use directly	77% missing
8	Fare is heavily right-skewed	Skew = 4.79 → use <code>log1p</code>
9	Embarkation port has a notable effect	Cherbourg: ~55% vs Southampton: ~34%
10	1st class females had near-perfect survival	96.8% survival rate

💡 Interview Q&A (Quick Reference)

Q1. What is EDA and why is it important? EDA is the process of analysing datasets to summarise their main characteristics — often visually — before formal modelling. It uncovers data quality issues, distributions, relationships, and hypotheses. Without it, models are built on poorly understood data.

Q2. Which plots do you use to check correlation? `sns.heatmap()` for correlation matrix, `sns.pairplot()` for pairwise relationships, and `sns.scatterplot()` for two-variable comparisons.

Q3. How do you handle skewed data? Use `log1p` for right-skewed data (like Fare), `sqrt` for moderate skew, or `Box-Cox` for general use. Tree-based models are also naturally robust to skew.

Q4. How to detect multicollinearity? Check the correlation heatmap for $|r| > 0.8$, and compute Variance Inflation Factor (VIF > 5-10 flags issues).

Q5. What are univariate, bivariate, and multivariate analyses? Univariate = one variable (histogram, boxplot). Bivariate = two variables (scatter, bar). Multivariate = three+ variables (pairplot, heatmap, facet grids).

Q6. Difference between heatmap and pairplot? Heatmap shows aggregated values (e.g., correlations) as a colour matrix — compact. Pairplot shows raw scatter plots for every variable pair + KDE on the diagonal — more detailed.

Q7. How do you summarize your insights? Identify impactful patterns, quantify with numbers, relate to domain context, flag data quality issues, and recommend next steps (imputation, transforms, feature engineering).

▶ How to Run

```
bash

# 1. Clone the repo
git clone https://github.com/your-username/task-5-titanic-eda.git
cd task-5-titanic-eda

# 2. Install dependencies
pip install pandas numpy matplotlib seaborn jupyter

# 3. Launch the notebook
jupyter notebook Titanic_EDA.ipynb
```

Note: Make sure `train.csv`, `test.csv`, and `gender_submission.csv` are in the same directory as the notebook. Dataset available at [Kaggle Titanic](#).

👤 Author

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